Consider an equation

$$Y'(t) + a(t) \cdot Y(t) = b(t) \cdot (y(t))^k \quad (k \in \mathbb{N})$$

Simply, $Y' + a(t) \cdot Y = b(t) \cdot y^k$

Let $Y = y^{1-k}$. Then, $Y' = (1-k) \cdot y^{-k} \cdot y'$

$$\frac{y'}{y^k} + a(t) \cdot y^{1-k} = b(t) \Rightarrow \frac{1}{1-k} \cdot Y' + a(t) \cdot Y = b(t)$$

$$\Rightarrow Y' + (1-k) \cdot a(t) \cdot Y = (1-k) \cdot b(t)$$

So, using integration factor, we can obtain a solution Ys, and then $y = Y^{\frac{1}{1-k}}$ is the solution.